Overview

In this study, we investigate the impact of having daughters on several outcomes.

Hypotheses

Hypothesis 1 treatment effect "attitudes":

Having daughters leads to stronger support for gender equality (as measured by attitudes).

Hypothesis 2 treatment effect "political preferences":

Having daughters leads to more left-leaning political preferences.

Hypothesis 3 treatment effect "gender-equal behaviors":

Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

Hypothesis 4 gendered treatment effect "attitudes":

Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

Sample

The data used will be SOEP data from the years 2000 to 2020 including the datasets pl, hl, biol, biobirth, bioparen, pgen, hgen, and ppathl. Access to data has been limited as authors accessed only 5% of the data prior to submitting the preregistration. The data used for the study ("SOEP") had been collected independently before. For the analysis we restrict ourselves to the household survey of the SOEP surveys, as we are interested in household decisions.

Variable overview

Outcome variables

- H1 & H4: Support for gender equality index, *support_index* (1 = low support, 4 = high support)
    - Here, we combine several measures of gender equality attitudes into one index. All measures were recorded in 2012 on a Likert-scale of 1 (fully agree) to 4 (do not agree at all). We recode opposite-framed items such that 1 = low support for gender equality and 4 = high support.
    - The following items are combined into the *support_index*:
        - Child below 6 years suffers if mom works [plh0298_v1]
        - Women Should Rather Care About Family Than Career [plh0301]
        - Child below 3 years suffers if mom works [plh0302_v1]
        - Men involved in housework [plh0303] – we reverse-coded this item
        - Men Care more About Financial Security [plh0307]
        - Working Mothers Have Equal Emotional Warmth (compared to non-working mothers) [plh0309] -- we reverse-coded this item

    - The decision which items to consider was based on theory, data comparability (same scale, same year) and on results from statistical tests (inter-item correlation as well as intra-item correlations, Cronbach's alpha).

- H2: Political orientation (0=extreme left, 10=extreme right), *plh0004* [plh0004]
- H3: Gender working hours gap on couple level (relative to total working hours per couple) i.e. the absolute difference in couple working hours divided by the sum of couple working hours, *diff_pgtatzeit* [pgtatzeit]

Treatment variable

Treatment *T1*:

We define being treated (the independent variable) as the first child of the respondent being a daughter (T1=1) vs. son (T1=0). We do this for various reasons. First, we believe that the first child has the biggest impact on our outcome variables. Second, if we would compare families with daughters to families without daughters, families without daughters would be very different in that they have less children (as the gender of the child is assumed to be random) and the control group would be much smaller than the treatment group.

Pre-/post-birth *post*:

We create a dummy for measures obtained pre- or post the birth of the first child.

Control variables

- Number of children per individual per year, num_kids [based on the variable kidgeb01, kidgeb02 and so on]
- Respondent gender (1=male, 2=female), *sex* [sex]
- Age when surveyed (in years), *bioage* [bioage], also used as Age^2 in our analysis
- Continent of birth (1 = Europe, 2 = Asia, 3 = Africa, 4 = North America, 5 = South America, 6 = Oceania, 7 = Unknown), *birth_continent* [transformed from matching the continents to corigin]
- Education (amount of education or training in years), *pgbilzeit* [pgen/pgbilzeit]
- Income (monthly household net income in EUR), *hghinc* [hgen/hghinc]
- Married when surveyed (0=no, 1=yes), *pgfamstd* [based on pgfamstd with all negative values being coded as missing and all but living together in an active marriage (=1) being coded as 0]
- Current residence in former West/East Germany (1=West, 2=East) *sampreg* [sampreg]
- Current Health (1=very good, 5=bad), *ple0008* [ple0008]

Missing data

In the SOEP, missing responses are coded with negative values and distinguished why they are missing. As for our analysis, the reasons for missing values do not matter, we recode all negative values for all original variables used from the SOEP to "." (missing). The same denotation is used for transformed and recoded variables.

Analysis plan

Hypothesis 1

Having daughters leads to stronger support for gender equality (as measured by attitudes).

Variables:

- Outcome variable: Support for gender equality index, *support_index* (1 = low support, 4 = high support)
- Treatment variable: *T1*
- Control variables as described above

Inclusion/Exclusion criteria

For the analysis we only use the year 2012, because in that year the questions used in our index were asked. We restrict our analyses to participants who have at least one child by 2012 at the time the survey was completed.

Exact analysis:

We use the following regression model specification to investigate treatment effects on attitudes (Hypothesis 1):

$$X_i = \beta_0 + \beta_1 T1 + \beta_2 C_i + \varepsilon$$

where X_i is participant *i*'s *support_index*, β_0 is the constant, β_1 captures the treatment effect and ε is the error term. C_i is the full set of control variables.

Interpretation of results

A positive coefficient of β_1 is indication for a support of the hypothesis (as higher values of the support index indicate higher support). We use a two-tailed test and a significance level of $\alpha = .05$.

Hypothesis 2

Having daughters leads to more left-leaning political preferences.

Variables:

- Outcome variable: Political orientation (0=extreme left, 10=extreme right), *plh0004* [plh0004]
- Treatment variable: *T1*, interaction with dummy *post*
- Control variables as described above

Inclusion/Exclusion criteria

We restrict the sample to participants whose first child is born between 2000 and 2020 as we don't observe them before 2000 and this would potentially influence our DiD-estimates.

Exact analysis:

We use the following regression model specification to investigate treatment effects on political preferences (Hypothesis 2):

$$Y_i = \alpha_G + \alpha_T + \beta_3 T1 * post + \beta_4 C2_i + \varepsilon_i$$

where Y_i is participant *i*'s political orientation [plh0004], α_G are the individual fixed effects and α_T are the time fixed effects and β_3 captures the interaction effect of having a daughter vs having a son as first child, and ε_i is the participant's error term (standard errors clustered on the individual level [*pid*]). $C2_i$ is the full set of time-variant control variables.

Interpretation of results

A negative coefficient of β_3 is indication for a support of the hypothesis (as lower values of political orientation indicate more left-leaning). We use a two-tailed test and a significance level of $\alpha = .05$.

Hypothesis 3

Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

Variables:

- Outcome variable: Gender working hours gap on a couple level (relative to total working hours per couple), *diff_pgtatzeit* [pgtatzeit]
- Treatment variable: *T1*, interaction with dummy *post*
- Control variables as described above plus a generated *coupleid* which uniquely identifies couples to cluster standard errors on the couple-level.

Inclusion/Exclusion criteria

We restrict the sample to participants whose first child is born between 2000 and 2020 as we don't observe them before 2000 and this would influence our DiD-estimates.

For this hypothesis, we exclude same-sex couples as we cannot compute our outcome variable, a gendered difference in their working hours, for those participants. Further, we only include working age population to ensure we can compute working hour differences. And as we look at a couple outcome we collapse the data to individual couples and for variables that differ within the couple we take the female variables (for simplicity as we are not interested in the fixed effects).

Exact analysis:

We use the following regression model specification to investigate treatment effects on gender-equal behaviors (Hypothesis 3):

$$Z_i = \alpha_G + \alpha_T + \beta_3 T1 * post + \beta_4 C3_i + \varepsilon_c$$

where Z_i is participant *i*'s gender working hours gap *[diff_pgtatzeit],* α_G are the individual fixed effects and α_T are the time fixed effects and β_3 captures the interaction effect of having a daughter vs having a son as first child, and ε_c is the participant's error term (standard errors clustered on the couple level [*coupleid*]). $C3_i$ is the full set of time-variant control variables by couple.

Interpretation of results

A negative coefficient of β_3 is indication for a support of the hypothesis (as lower values indicate a smaller gender working hours gap). We use a two-tailed test and a significance level of α =.05.

Hypothesis 4

Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

Variables:

- Outcome variable: Support for gender equality index, *support_index* (1 = low support, 4 = high support)
- Treatment variable: *T1,* interaction with *sex*
- Control variables as described above but excluding sex

Inclusion/Exclusion criteria

For the analysis we only use the year 2012, because in that year the questions used in our index were asked. We restrict our analyses to participants who have at least one child by 2012 at the time the survey was completed.

Exact analysis:

We use the following regression model specification to investigate gendered treatment effects on attitudes (Hypothesis 4):

$$X_i = \beta_0 + \beta_1 T1 + \beta_2 sex + \beta_3 T1 * sex + \beta_4 C4_i + \varepsilon$$

where X_i is participant *i*'s *support_index*, β_0 is the constant, β_1 captures the effect of having a daughter as a first child compared to a son, β_2 captures the effect of the parent's sex (with female = 1 and male = 0) and β_3 captures the gendered treatment effect and ε is the error term. $C4_i$ is the limited set of control variables (excluding sex).

Interpretation of results

A negative coefficient of β_3 is indication for a support of the hypothesis (as lower values for women indicate a stronger effect for men, assuming that β_1 is positive). We use a two-tailed test and a significance level of α =.05.